

Model	Method	Cancer-Myth		CREPE		Medical QA			General QA	
		FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	MQ ↑	PQ ↑	MB ↑	MMLU ↑	GSM ↑
Qwen2.5 7B	Plain	2.4 ^H	100.0 ^H	—	—	—	—	—	—	—
	Always correct	56.9 ^H	21.5 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	CoT	0.7 ^H	99.3 ^H	—	—	—	—	—	—	—
	Review CoT (2-step)	10.5 ^H	94.6 ^H	—	—	—	—	—	—	—
	PreWoMe-style	—	—	—	—	—	—	—	—	—
	Extract + Verify	—	—	—	—	—	—	—	—	—
	Direct-gated response	6.9 ^H	90.6 ^H	—	—	—	—	—	—	—
	CoT-gated response	21.3 ^H	85.9 ^H	—	—	—	—	—	—	—
	Text-gated response	50.7 ^H	60.4 ^H	—	—	—	—	—	—	—
	Extract+Verify (scope)	5.7 ^H	99.3 ^H	—	—	—	—	—	—	—
	Self-gated FP identification	7.0 ^H	89.9 ^H	—	—	—	—	—	—	—
	Probe-gated response	44.8 ^H	64.4 ^H	—	—	—	—	—	—	—
	Plain (LoRA cohort)	1.7 ^T	100.0 ^T	—	—	—	—	—	—	—
	DPO (FPQ + twins)	2.6 ^T	100.0 ^T	—	—	—	—	—	—	—
	SFT (FPQ + twins)	29.1 ^T	45.6 ^T	—	—	—	—	—	—	—
	GEPA	—	—	—	—	—	—	—	—	—
	Ours	—	—	—	—	—	—	—	—	—
GPT-6 Luna	Plain	54.5 ^L	91.0 ^L	72.7 ^R	98.9 ^R	—	—	—	—	—
	Always correct	—	—	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	CoT	—	—	—	—	—	—	—	—	—
	Review CoT (2-step)	—	—	—	—	—	—	—	—	—
	PreWoMe-style	88.9 ^L	59.0 ^L	—	—	—	—	—	—	—
	Extract + Verify	87.9 ^L	48.0 ^L	—	—	—	—	—	—	—
	Direct-gated response	—	—	—	—	—	—	—	—	—
	CoT-gated response	—	—	—	—	—	—	—	—	—
	Text-gated response	—	—	—	—	—	—	—	—	—
	GEPA	78.8 ^L	72.0 ^L	74.7 ^R	98.0 ^R	—	—	—	—	—
	Ours	—	—	—	—	—	—	—	—	—
Gemma 4 12B	Plain	4.3 ^H	100.0 ^H	—	—	—	—	—	—	—
	Always correct	72.0 ^H	2.0 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	CoT	3.4 ^H	99.3 ^H	—	—	—	—	—	—	—
	Review CoT (2-step)	41.2 ^H	78.5 ^H	—	—	—	—	—	—	—
	PreWoMe-style	—	—	—	—	—	—	—	—	—
	Extract + Verify	—	—	—	—	—	—	—	—	—
	Direct-gated response	37.2 ^H	83.9 ^H	—	—	—	—	—	—	—
	CoT-gated response	43.4 ^H	67.1 ^H	—	—	—	—	—	—	—
	Text-gated response	63.8 ^H	50.3 ^H	—	—	—	—	—	—	—
	Probe-gated response	—	—	—	—	—	—	—	—	—
	GEPA	—	—	—	—	—	—	—	—	—
	Ours	—	—	—	—	—	—	—	—	—
Qwen3.8 27B OFF	Plain	49.7 ^H	79.9 ^H	—	—	—	—	—	—	—
	Always correct	71.0 ^H	28.9 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	CoT	54.7 ^H	71.1 ^H	—	—	—	—	—	—	—
	Review CoT (2-step)	61.4 ^H	55.0 ^H	—	—	—	—	—	—	—
	PreWoMe-style	—	—	—	—	—	—	—	—	—
	Extract + Verify	—	—	—	—	—	—	—	—	—
	Direct-gated response	53.0 ^H	72.5 ^H	—	—	—	—	—	—	—
	CoT-gated response	62.1 ^H	59.7 ^H	—	—	—	—	—	—	—
	Text-gated response	68.3 ^H	49.0 ^H	—	—	—	—	—	—	—
	Probe-gated response	—	—	—	—	—	—	—	—	—
	GEPA	—	—	—	—	—	—	—	—	—
	Ours	—	—	—	—	—	—	—	—	—
Qwen3.8 27B ON	Plain	40.7 ^H	88.6 ^H	—	—	—	—	—	—	—
	Always correct	83.4 ^H	5.4 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	CoT	43.1 ^H	87.9 ^H	—	—	—	—	—	—	—
	Review CoT (2-step)	66.7 ^H	57.0 ^H	—	—	—	—	—	—	—
	PreWoMe-style	—	—	—	—	—	—	—	—	—
	Extract + Verify	—	—	—	—	—	—	—	—	—
	Direct-gated response	53.5 ^H	75.8 ^H	—	—	—	—	—	—	—
	CoT-gated response	62.1 ^H	60.4 ^H	—	—	—	—	—	—	—
	Text-gated response	78.2 ^H	48.3 ^H	—	—	—	—	—	—	—
	Probe-gated response	—	—	—	—	—	—	—	—	—
	GEPA	—	—	—	—	—	—	—	—	—
	Ours	—	—	—	—	—	—	—	—	—

Table 2. Response quality and preservation of general QA performance

Stored results; — means unavailable or not comparable. Values are percentages, one run.
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.
FPQ / NFP / TPQ: Well ≥ 4. MQ / PQ / MB: MedQA / PubMedQA / Medbullets.
L: Luna matched 99 FPQ / 100 NFP, exposed test; R: CREPE 751 FPQ / 2253 TPQ.
T: legacy 234 FPQ / 149 natural NFP; synthetic twins were used in training.
Gate rows route saved Plain/correction answers; not exact Two Axes reproduction.
Ours is not fixed to SAE. Source counts and comparison limits: result_sources.md.